Artificial Intelligence and Digital Technologies in Electrical Construction: A Systematic Review of Applications, Maturity, and Research Directions

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ABSTRACT

The electrical construction industry faces persistent challenges in coordination, productivity, safety, and workforce development, and artificial intelligence (AI) is widely proposed as a remedy. Evidence specific to the electrical trade, however, remains fragmented across studies that treat electrical work as one component of broader mechanical, electrical, and plumbing (MEP) systems. This review synthesizes 129 peer-reviewed studies published between 2008 and 2026, retrieved from Scopus and Web of Science and coded across six dimensions: AI technique, digital technology, construction phase, electrical work category, maturity level, and industry involvement. Building information modelling dominates, appearing in approximately half of the studies, while large language models and generative AI remain scarce. Research concentrates on preconstruction coordination and closeout documentation, with limited attention to execution-phase productivity, code compliance automation, and cost estimation. Publication volume more than tripled after 2021 while the field-tested share fell from above 75% to between 29% and 45%, and quantified benefits are dominated by algorithm performance rather than by cost, safety, or productivity outcomes. The review proposes a taxonomy organized by the electrical contractor value chain and sets out six priority research directions, including LLM-assisted code compliance and AI-driven cost estimation.

Keywords: Artificial intelligence; machine learning; deep learning; large language models; electrical contractors; building information modelling (BIM)

INTRODUCTION

The electrical construction industry continues to face well-documented challenges in jobsite coordination, labor productivity, worker safety, and workforce development (Barbosa et al., 2017). Productivity in construction has stagnated relative to other industries, and electrical contractors bear a disproportionate share of coordination-related delays and rework costs (Gholizadeh et al., 2021). Artificial intelligence (AI) and digital technologies have increasingly been proposed to address these inefficiencies, in applications ranging from automated scheduling and clash detection to safety monitoring and workforce training (Pan & Zhang, 2021). Much of this research, however, does not treat electrical work as a distinct trade. Electrical systems are instead studied as part of broader mechanical, electrical, and plumbing (MEP) bundles, which dilutes findings and makes them difficult to apply directly to electrical contractors (J. Wang et al., 2016). The evidence

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base specific to electrical construction therefore remains fragmented, leaving practitioners little research to guide technology adoption.

This paper addresses that gap through a systematic review of AI and digital technologies in electrical construction, drawing on 129 peer-reviewed studies published between 2008 and 2026 and retrieved from Scopus and Web of Science. Studies were screened against predefined relevance criteria and coded across six dimensions: AI technique, digital technology, construction phase, electrical work category, maturity level, and industry involvement; Figure 1 summarizes the selection process. Existing broad reviews of AI in construction have mapped general trends and techniques (Abioye et al., 2021; Pan & Zhang, 2021), but none applies a taxonomy grounded in the electrical contractor value chain or assesses maturity and industry relevance at the trade level. This review makes three contributions. It establishes such a taxonomy; it characterizes the maturity, funding, and industry-engagement profile of the evidence base across six analytical views; and it derives a forward research agenda from the reviewed studies' own stated future directions rather than from the authors' priorities alone.

RESEARCH METHODOLOGY AND DESIGN

This review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al., 2022). Because the retrieved evidence is heterogeneous in design, unit of analysis, and reported outcomes, and therefore does not support statistical pooling, the synthesis additionally follows the Synthesis Without Meta-analysis (SWiM) guideline (Campbell et al., 2020), which requires grouping decisions, the coding scheme, and the basis for prioritizing findings to be stated in advance. Together they keep screening PRISMA-conformant while making the synthesis logic explicit and auditable.

The four-stage process of identification, screening, eligibility assessment, and inclusion, together with the number of records retained at each stage, is summarized in Figure 1.

Identification and Screening

Scopus and Web of Science were searched on 23 April 2026 using keywords relevant to AI and digital technology in electrical construction; both databases were selected for their coverage of peer-reviewed journals and their capacity to export structured bibliometric data. The search was applied to titles, abstracts, and keywords and combined three term blocks: terms identifying the electrical trade; terms identifying AI and digital technologies, joined with AND; and terms for non-construction electrical contexts such as power grids, electric vehicles, and photovoltaics, excluded with AND NOT. The full search string is given in Appendix A (Table A1). The search covered publications from 2006 to the search date and returned 1,085 records, of which 535 English-language journal articles and reviews proceeded to title and abstract screening. The earliest retained study dates from 2008. Because the search closed on 23 April 2026, 2026 is an incomplete publication year: the 11 included studies from 2026 represent less than four months of indexing, and the 2026 values in Figures 2 and 3 understate annual output.

Each title and abstract was screened against two inclusion criteria, relevance to electrical construction and application of AI or digital technology to a construction problem, and against six exclusion criteria: medical use of the "MEP" acronym, power system operations unrelated to construction, non-peer-reviewed publications, unrelated electrical research, other unrelated uses of the acronym, and papers lacking substantive electrical construction application. Each record received a judgment on both dimensions with a written justification, leaving 172 records for full-

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text review. Studies that address MEP systems generically were retained where electrical systems formed part of the modeled, coordinated, or inspected scope. Each retained study was then coded for electrical scope as electrical-only (n = 36), MEP-wide (n = 89), or building-wide (n = 4), so that trade-specific evidence remains identifiable within the MEP aggregate.

Eligibility Assessment and Inclusion

The 172 retained records underwent full-text review and structured coding across forty-four dimensions, spanning research problem and method, AI technique, digital technology, construction phase, electrical work category, maturity level, contribution type, pain point addressed, contractor role impacted, code compliance, industry-body affiliation, data availability, and acknowledged limitations. This common frame is the standardized scheme through which studies of differing design are made comparable, as SWiM requires when no common effect measure is available.

Coding followed a written codebook that fixed, before extraction began, the permitted categories and decision rules for every dimension. One author verified the codes of all 129 included studies against the full text, and borderline cases were reviewed by the second author and resolved by discussion. AI technique and digital technology were coded as separate dimensions. An AI technique was recorded only where a study applied a learning, reasoning, or optimization method, namely classical machine learning, neural networks, deep learning, computer vision, natural language processing, large language models, expert or rule-based systems, fuzzy logic, evolutionary algorithms, or statistical modeling. BIM, digital twins, laser scanning and point clouds, extended reality, IoT sensing, wearables, and robotics were recorded as enabling digital technologies and were not counted as AI. On this basis, 56 of the 129 studies (43.4%) apply an AI technique and 73 (56.6%) apply digital technologies without an AI component.

Maturity level records how far the validation of a contribution was taken within the paper and was coded on a three-level ordinal scale. Conceptual studies propose an idea, framework, or model without implementation or empirical test. Lab-validated studies implement the contribution and test it under controlled conditions, using synthetic, benchmark, or limited project data. Field-tested studies validate the contribution on a live project or in an operating industry setting. The scale describes the strength of validation evidence and not the magnitude of benefit, since a field-tested study shows that a tool functioned on a project, not that the project improved. Maturity is therefore read together with the benefit each study quantifies, classified on four tiers: technical capability (algorithm or model performance, such as accuracy or mean intersection over union), operational effectiveness (task-level time, workflow, coordination, or training outcomes), project outcomes (cost, labor productivity, safety, or quality measured on a project), and organizational value (firm-level adoption, sustained use, or business performance). Taken together, the two dimensions describe a progression from concept development, through prototype validation and field implementation, to measurable project outcomes, organizational adoption, and wider industry impact. Among the reviewed studies, quantified evidence stops at the project-outcome tier.

Forty-three records were excluded at this stage for marginal electrical relevance on close reading, tangential or absent AI application, misleading abstracts, retraction or withdrawal, publication in outlets failing established indicators of scholarly quality, or duplication with included journal articles. The remaining 129 studies form the analytic base for the descriptive overview, the thematic synthesis, and the identification of underserved combinations.

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Data Extraction and Analysis

The coding matrix was analyzed in three passes: a descriptive pass summarizing the distribution of studies across publication year, geography, method type, AI technique, digital technology, construction phase, and maturity level; a thematic synthesis grouping studies by pain point and claimed value proposition to characterize what the evidence supports and where claims outrun the data; and a cross-tabulation of AI technique against construction phase, and electrical work category against method type, to locate the sparse cells reported here as underserved combinations.

Groups were formed on coded dimensions rather than reported effect size, and no study was weighted by sample size, journal, or citation count. Quantified benefits are attributed to the individual study and its stated context rather than aggregated, because the underlying outcomes are measured on incommensurable scales and the field-tested base is thin. Certainty therefore rests on the density and consistency of coded evidence within a group and is described qualitatively.

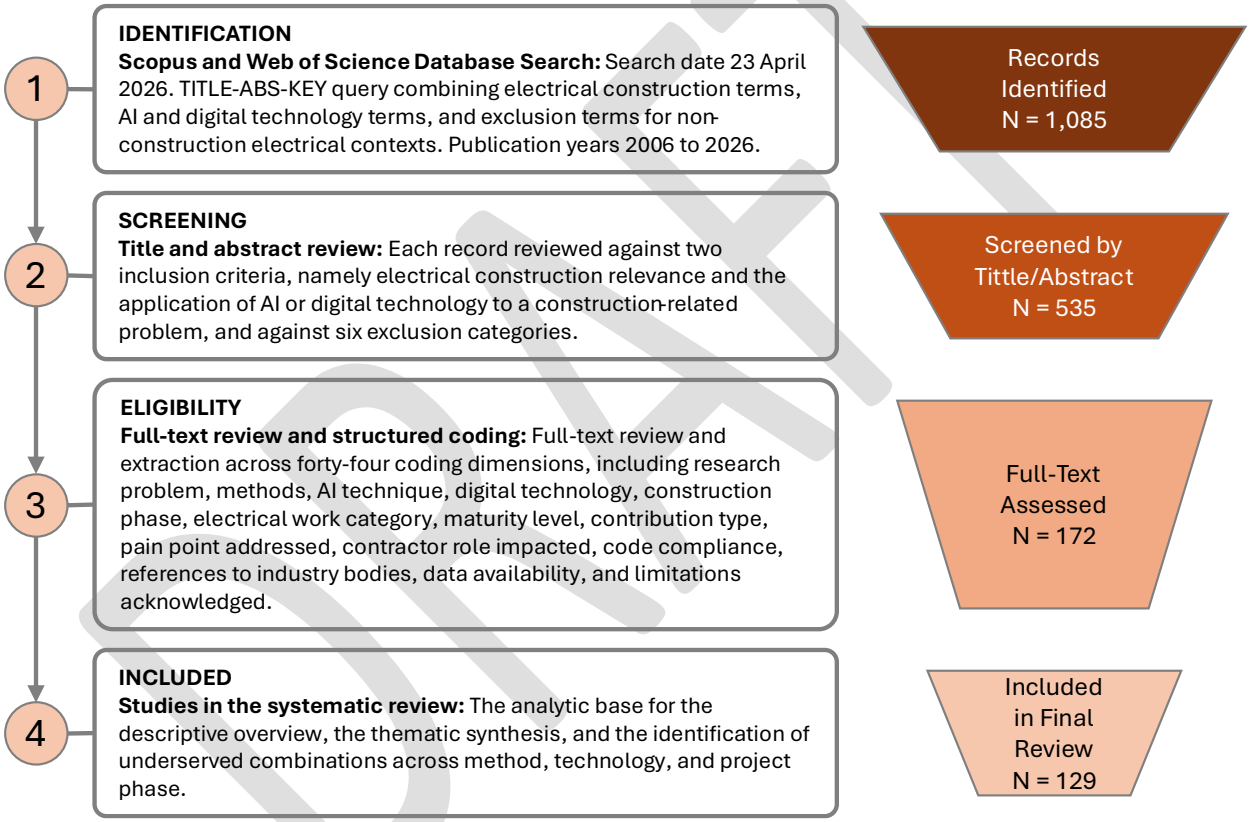


Figure 1. PRISMA Chart

RESULTS

Publication Growth and the Erosion of Field-Testing

Figure 2 shows a field expanding rapidly while shifting in how knowledge is produced and validated. From 2008 through roughly 2020, annual output rarely exceeded eight papers, yet the share of field-tested work hovered near or above 75%. Early contributions were grounded in live project conditions: Khanzode et al. (2008) documented labor savings of 20 to 30% and near-zero rework on a $96.9M healthcare project using BIM-led MEP coordination, while Dossick and Neff

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(2010) drew on more than twelve months of ethnographic observation to show that technological integration alone does not produce organizational cohesion.

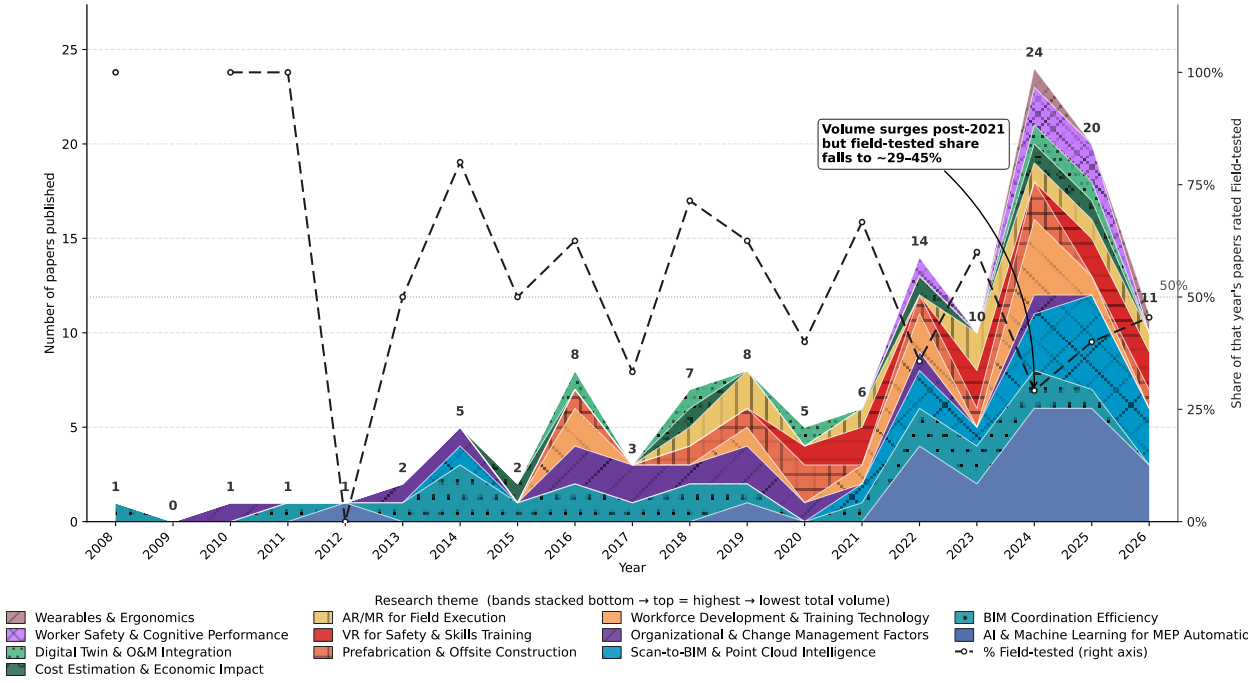


Figure 2. Publication Trends in Construction Technology Research by Theme and Field-Testing Rate, 2008-2026

The volume surge beginning in 2021, peaking at 24 papers in 2024, tells a more complicated story. Publication counts more than tripled relative to the preceding decade, but the field-tested share collapsed to roughly 29 to 45%. This inverse relationship between quantity and real-world validation suggests that growth is driven substantially by laboratory experiments and simulations rather than by deployment in functioning construction environments. Jing et al. (2024), for example, achieved semantic segmentation accuracy of 97.39% on MEP point clouds, but under lab-validated conditions using curated datasets, raising legitimate questions about performance in the cluttered, dynamic, poorly lit environments typical of active sites.

The thematic composition of the stacked bands also shifts after 2021. Scan-to-BIM and Point Cloud Intelligence, AI and Machine Learning for MEP Automation, and Digital Twin and O&M Integration occupy increasingly large portions of the post-2021 stack, all computationally intensive domains in which controlled experimental setups are easier to manage and publish than longitudinal field deployments. Wang et al. (2021) are a partial exception, validating a fully automated laser-scanning-to-BIM pipeline across three Hong Kong sites, but such bridging work remains a small cohort.

The decline matters because construction is among the least forgiving environments for algorithmic failure. Park and Lee (2017) found that BIM-led coordination delivered 228% faster productivity than BIM-assisted coordination on a hospital project, a result possible only because two parallel real-world towers were observed under comparable conditions. Lee and Kim (2014) similarly showed that sequential versus parallel coordination strategy produced measurable

differences in clash detection and meeting efficiency. Such findings carry a credibility lab benchmarks cannot replicate.

Thematic gaps are also visible. Cost Estimation and Economic Impact, Organizational and Change Management Factors, and Workforce Development appear as thin bands throughout, underrepresented relative to their practical relevance. Jang and Lee (2018) is one of the few field studies examining the economic and productivity trade-offs of BIM-based multi-trade prefabrication, finding a 40% reduction in assembly person-hours through learning effects, yet this line of inquiry has not attracted the follow-on volume that computational themes have. Alshboul and Shehadeh (2026) offer a recent field-tested counterexample in AI-driven clash detection, but one paper is not a trend. The chart registers a field that is technically accelerating but empirically retreating.

Shifting Contribution Types and Research Maturity Over Time

Figure 3 shows a field that has grown dramatically in volume while remaining structurally uneven in contribution type and research maturity. From 2008 to approximately 2019, publication activity was modest and stable, anchored by empirical findings and best practices emerging from early BIM adoption in MEP coordination. Foundational work from this period, such as Khanzode et al. (2008) and Dossick and Neff (2010), was predominantly field-tested, reflecting a time when researchers engaged directly with construction sites. That early grounding is what the field has struggled to maintain as it scaled.

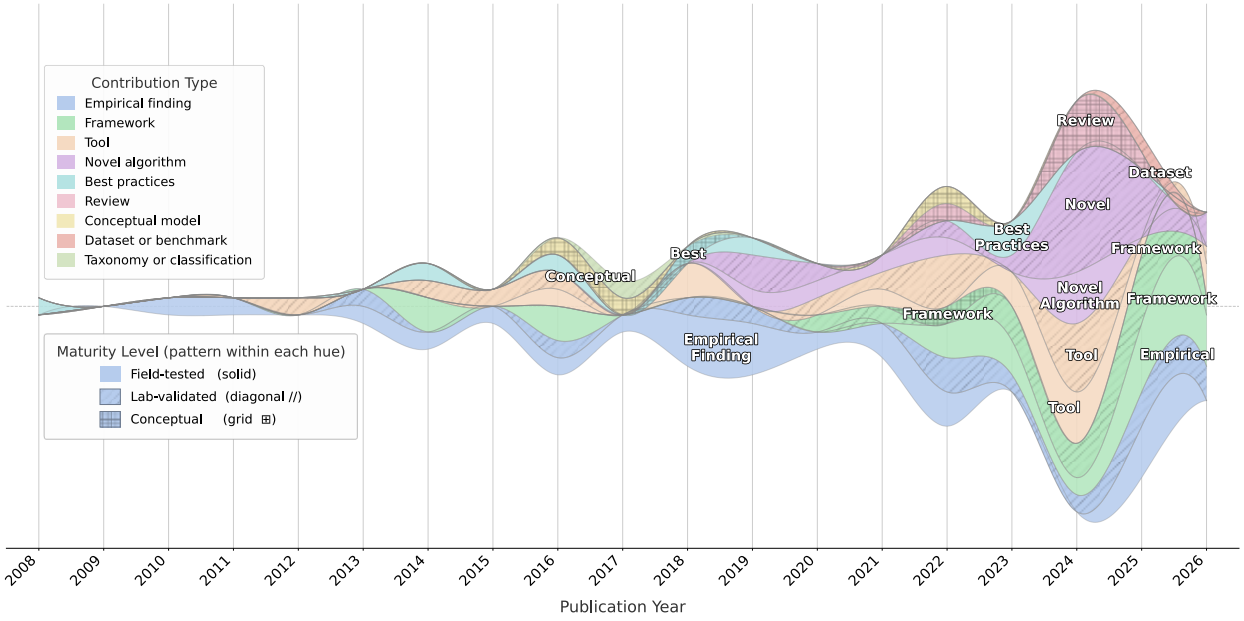


Figure 3. Evolution of research contributions by type and maturity level across the reviewed literature (2008-2026)

Between 2013 and 2019 the streamgraph shows a marked expansion of conceptual models and frameworks, coinciding with growing interest in automation and BIM-integrated workflows. Akponeware and Adamu (2017) is illustrative: a conceptual contribution identifying systemic coordination problems without offering a validated solution pathway. Such work is necessary to frame emerging problems, but conceptual outputs accumulated faster than field-tested ones during this period, widening the gap between problem identification and demonstrated resolution.

The most striking pattern is the steep surge beginning around 2022 and accelerating through 2026, correlating with the proliferation of deep learning and, more recently, large language model approaches in construction. Li et al. (2026) and Wang et al. (2024) exemplify this wave with sophisticated frameworks for automated constructability checking and Scan-to-BIM reconstruction. The maturity encoding should nonetheless give pause: a substantial proportion of recent contributions remain lab-validated, meaning claimed accuracy rates and efficiency gains are measured in controlled settings that may not reflect the variability of live construction.

The surge in review papers between 2024 and 2025 is both expected and concerning. Systematic reviews such as Yan et al. (2024) consolidate a literature expanding faster than any single practitioner can track, yet a field whose output increasingly consists of secondary literature reviewing its own prior output risks a self-referential loop rather than new empirical knowledge. The growth in dataset and benchmark contributions toward 2026 is more encouraging, suggesting the community is building the evaluation infrastructure that replicable progress requires.

What Figure 3 ultimately exposes is a trajectory shaped more by technological opportunity than by systematic identification of industry need. The best-practice and empirical contributions that dominated the early period, and that produced the most practically significant results, have not kept pace with algorithmic novelty. As frameworks multiply and algorithms are refined in laboratory conditions, accumulated knowledge of how organizational dynamics, contractual structures, and site-level variability interact with digital tools (Dossick & Neff, 2010; Pärn et al., 2018) risks being sidelined. Volume without depth of validation does not constitute maturity.

BIM as the Central Node: Mapping Conceptual Clusters

The keyword co-occurrence analysis (Figure 4), conducted in VOSviewer, reveals a field organized around a few dominant anchors rather than a well-distributed research landscape. Building Information Modeling (BIM) is the unambiguous center of the network, connecting to virtually every other cluster and functioning as the shared infrastructural language of the field, a role evident as far back as Khanzode et al. (2008). That dominance may equally reflect publication bias toward BIM-centric contexts, and BIM's persistent coupling with long-established terms such as Clash Detection, CAD, and IFC raises the question of whether the field is applying modern AI to genuinely new problems or retrofitting it to existing workflows.

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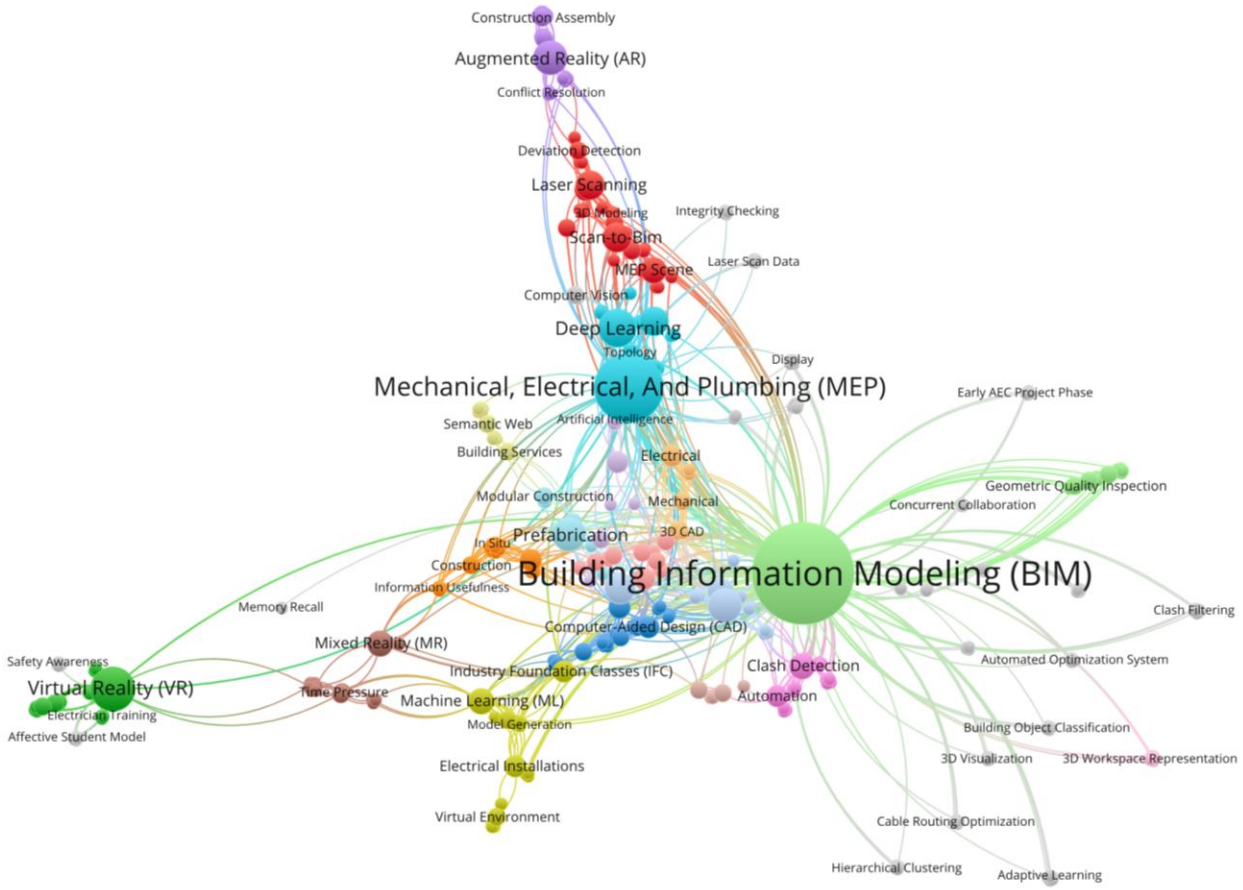


Figure 4. Keyword co-occurrence network of AI and immersive technologies in MEP-BIM research (2008-2026)

The second most prominent cluster coalesces around Deep Learning, Computer Vision, and Machine Learning, strongly linked to the Laser Scanning-Scan-to-BIM axis, with the most technically ambitious contributions focused on point cloud processing. Jing et al. (2024) achieve a mIoU of 90.76% for MEP semantic segmentation, and Wang et al. (2021) fully automate the scan-to-BIM pipeline for MEP scenes. Classical machine learning clusters separately around Clash Detection and Automation, exemplified by Alshboul and Shehadeh's (2026) XGBoost-based real-time system. LLM, Generative AI, and Foundation Models are absent from the map entirely, revealing a temporal lag relative to current AI development.

The AR, VR, and MR cluster links to BIM through 3D visualization and field inspection workflows. Chalhoub et al. (2021) temper optimistic AR narratives, finding it effective for large deviations but unreliable below two-inch tolerances, and Girgin et al. (2023) report that MR can halve MEP issue resolution time, though from a single case study. VR sits at the periphery, connected mainly to training and memory recall terms (Astaneh Asl et al., 2023), suggesting that educational applications remain underinvested.

Several structural gaps warrant attention. The hub-and-spoke topology indicates that research communities across AR, deep learning, and prefabrication work in parallel rather than converging, despite clear technical synergies. Organizational and behavioral terms sit at the periphery, a concern Dossick and Neff (2010) raised early by showing that BIM projects can be tightly coupled technologically yet divided organizationally, a finding that has generated little

follow-on research. Most critically, sustainability, carbon, and energy performance terms are almost entirely absent, a significant misalignment with the decarbonization imperatives now facing the sector, given that MEP systems govern most operational building energy consumption.

AI Technique Distribution Across Construction Phases

Figure 5 maps the intersection of AI technique and construction phase across the reviewed literature, with bubble size proportional to the number of papers and color encoding maturity level (conceptual, lab-validated, or field-tested). The figure is restricted to studies that apply an AI technique. Of the 129 studies, 56 (43.4%) do so, while 73 (56.6%) rely on BIM and other digital technologies without an AI component, and the two groups differ in maturity: 17 of the 56 AI studies (30.4%) are field-tested, against 45 of the 73 digital-technology studies (61.6%). The field-tested evidence base therefore rests largely on BIM-enabled practice, not on AI.

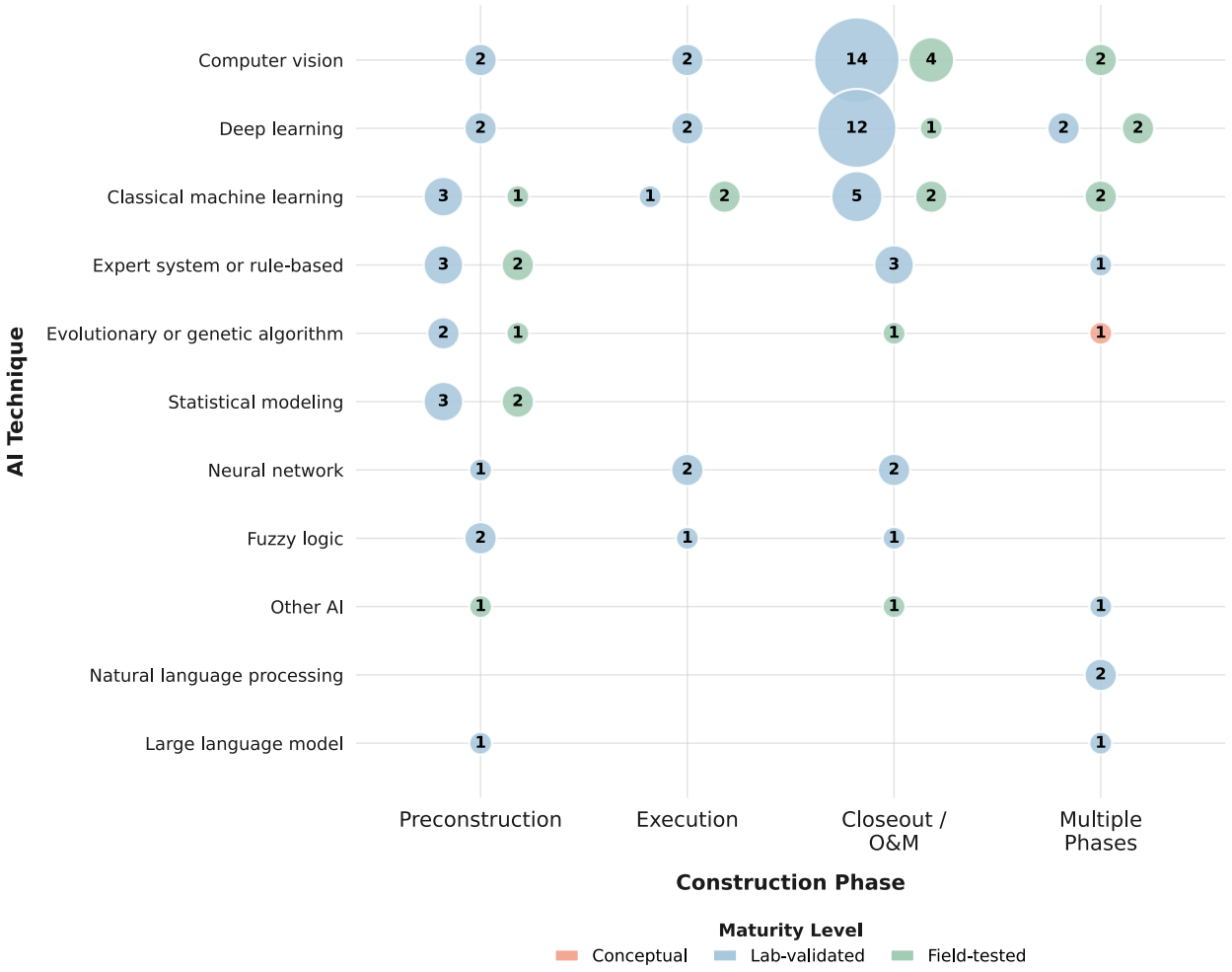


Figure 5. Distribution of AI Techniques Across Construction Phases and Maturity Levels in MEP-Related Research

The most striking feature is the concentration of computer vision and deep learning papers in the Closeout/O&M phase: 14 lab-validated and 4 field-tested for computer vision, and 12 lab-validated and 1 field-tested for deep learning. This reflects a well-defined trajectory focused on automated as-built documentation and scan-to-BIM workflows for MEP systems (Jing et al., 2024; Son et al.,

2019; B. Wang et al., 2021, 2024, 2025). Technically sophisticated as this work is, it weights research attention toward a single phase at the expense of phases where errors are still preventable rather than merely recorded. Most of these papers also remain lab-validated, reporting strong benchmark metrics from controlled, single-site environments with curated datasets, conditions far removed from the dust, occlusion, and multi-trade congestion of active sites.

The Preconstruction phase shows a more diverse spread of techniques, including classical machine learning, expert systems, statistical modeling, and, most recently, large language models, but most cells contain only one to three papers. Statistical modeling and fuzzy logic appear exclusively at this phase, a narrow niche in early cost estimation that has not expanded downstream. Expert systems achieve a comparatively favorable field-tested ratio (Li et al., 2026; Lu & Wong, 2018), arguably underappreciated given their interpretability and alignment with codified specifications.

Most revealing is the scarcity of research targeting the Execution phase, where on-site coordination, sequencing, and quality inspection occur and where many MEP errors originate. Shamsollahi et al. (2024) and Alshboul and Shehadeh (2026) are isolated contributions rather than a coherent research thread. Across the chart, the dominance of lab-validated over field-tested evidence is systemic: field-tested results are limited to selected classical machine learning (Alshboul & Shehadeh, 2026), expert systems (Lu & Wong, 2018), and a few computer vision applications (B. Wang et al., 2021, 2025). Studies achieving high accuracy on curated datasets rarely report deployment cost, integration difficulty, or performance degradation in production.

Natural language processing and large language models together account for only four papers, all lab-validated. Given the rapid adoption of LLM-based tools in adjacent industries and their potential for automating specification parsing and compliance checking, this scarcity likely reflects publication lag, but it leaves practitioners with little empirical guidance. The multi-agent framework of Li et al. (2026), noteworthy for 92% accuracy in constructability violation detection, remains a proof of concept validated on few projects. Figure 5 thus reveals a literature that is technically maturing but phase-skewed, maturity-stalled, and still nascent in its engagement with the techniques transforming adjacent industries.

Tracing Research Impact: From Funding Source to Quantified Benefit

Figure 6 reveals a literature landscape that, while growing in volume, exhibits several structural tensions that collectively raise questions about the field's real-world relevance and research transparency.

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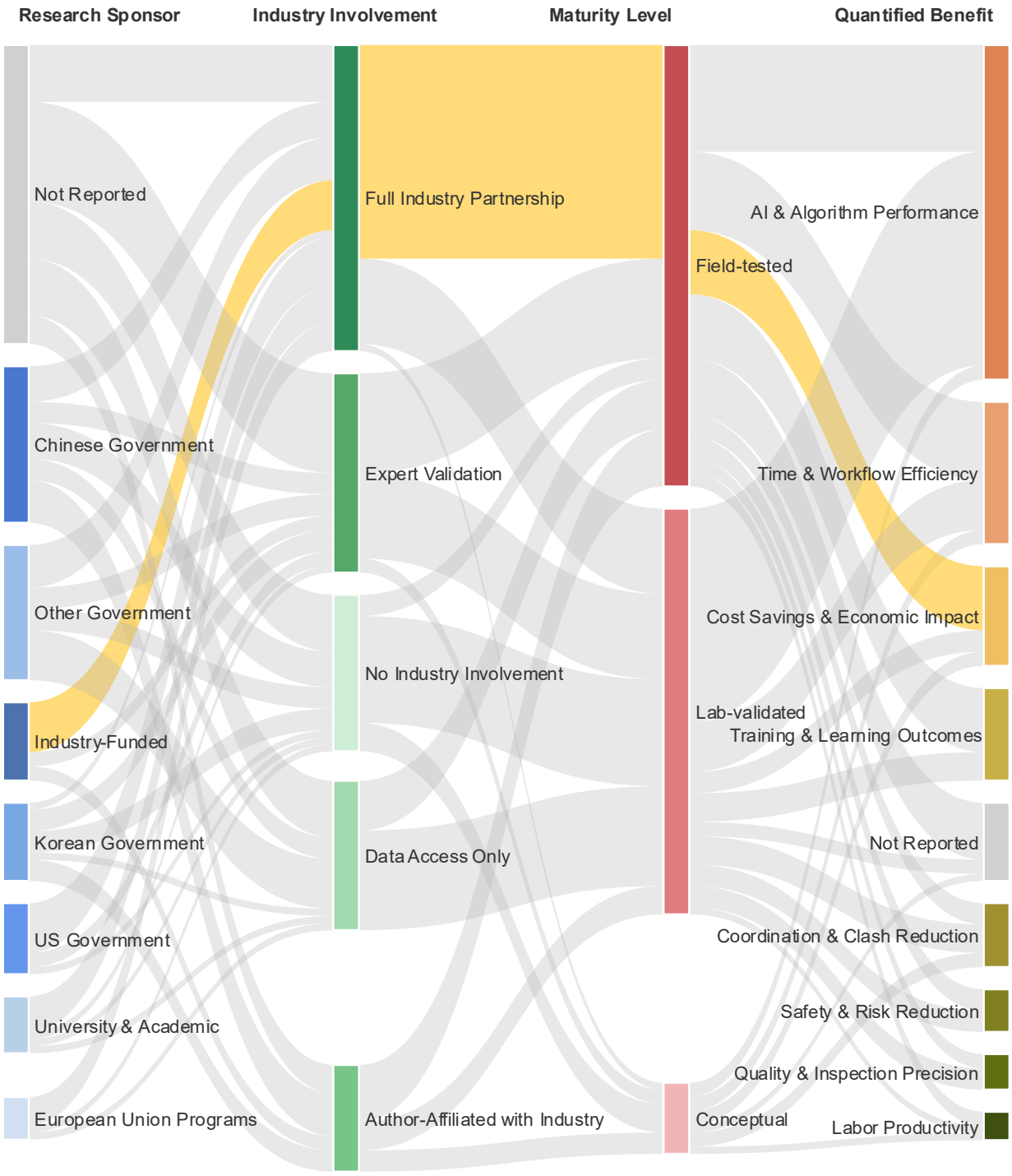


Figure 6. Sankey diagram mapping research sponsor, industry involvement, maturity level, and quantified benefit across the reviewed literature (n = 129)

The largest funding category at the leftmost node is "Not Reported" (n = 42, approximately 33% of the reviewed studies). This opacity is not trivial: funding source shapes research design, site access, and which outcomes get measured. Government agencies dominate the reported sponsors,

with Chinese Government (22), Other Government (19), Korean Government (11), U.S. Government (10), and European Union programs (6) accounting for most traceable funding. Industry-funded research represents only 11 papers, a small fraction for a domain positioned as an applied technology field, suggesting agendas driven more by academic and policy priorities than by contractor needs.

The industry involvement column is similarly mixed. Full Industry Partnership (43) is the largest category, but read alongside No Industry Involvement (22), Data Access Only (21), Author-Affiliated with Industry (15), and Conceptual (10), roughly half the literature operates without genuine field engagement. Data access alone, where industry supplies datasets but does not participate in study design or validation, is not equivalent to partnership. Dossick and Neff (2010) cautioned that technology adoption in construction is mediated by organizational and contractual structures that cannot be understood from the outside; the prevalence of arms-length data-sharing here suggests the lesson has not been absorbed. The maturity column appears balanced, Field-tested (62) against Lab-validated (57), but the framing is generous: lab validation shows that a tool functions under controlled conditions, not that it performs amid the noise, schedule pressure, and coordination complexity of a live site.

The sharpest critique emerges at the rightmost node. "AI & Algorithm Performance" dominates quantified benefits at 47 papers, more than double any other category. That metric captures whether a model achieves a competitive accuracy score; it does not capture whether deploying the model saves money, reduces injuries, or shortens schedules. Cost Savings & Economic Impact (14), Safety & Risk Reduction (6), Quality & Inspection Precision (5), and Labor Productivity (4) trail well behind. Studies that do reach the cost-savings node, such as Khanzode et al. (2008) with labor savings of 20-30% and roughly $9M in project cost reduction, show that economically meaningful outcomes are achievable, but they remain exceptions. Eleven papers quantified no benefit at all. Placed on the four benefit tiers defined in the methodology, 47 studies (36.4%) evidence technical capability only, 42 (32.6%) operational effectiveness, and 29 (22.5%) project outcomes, while none quantifies organizational value. Only 16 studies (12.4%) are both field-tested and report a project-level outcome. Algorithm accuracy is accordingly treated in this review as evidence of technical capability and not as evidence of improved construction performance.

The diagram's highlighted gold path, Industry-Funded to Full Partnership to Field-tested to Cost Savings & Economic Impact, represents the ideal research trajectory, and the need to highlight it underscores how rare it is. Lu and Wong (2018) and Tao et al. (2024) exemplify industry-partnered, field-validated work with tangible outcomes, yet such studies form a thin sliver of the reviewed studies. A substantial body of literature demonstrates algorithmic novelty under favorable conditions while evidence of deployment-ready, cost-effective impact remains scarce.

Where the Evidence Concentrates: Pain Points, Findings, and Realized Benefit

Figure 7 pairs the problems the literature sets out to solve with the findings it reports (panel a) and the benefits it quantifies (panel b), exposing a mismatch between what electrical contractors struggle with and where research effort lands. In panel a, bubble mass concentrates in the upper-left quadrant, where coordination and design-automation pain points meet three finding themes: machine learning automates MEP design and detection (23 papers), BIM coordination cuts the time and cost of clash resolution (21 papers), and point clouds reconstruct as-built conditions (16 papers). These three columns absorb roughly half of the reviewed studies, populated by Alshboul and Shehadeh (2026), Li et al. (2026), Khanzode et al. (2008), and Bosché et al. (2014). The

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coordination problem is well served, but the density reflects convergence on a narrow band of tractable, model-centric problems rather than the breadth of contractor pain points.

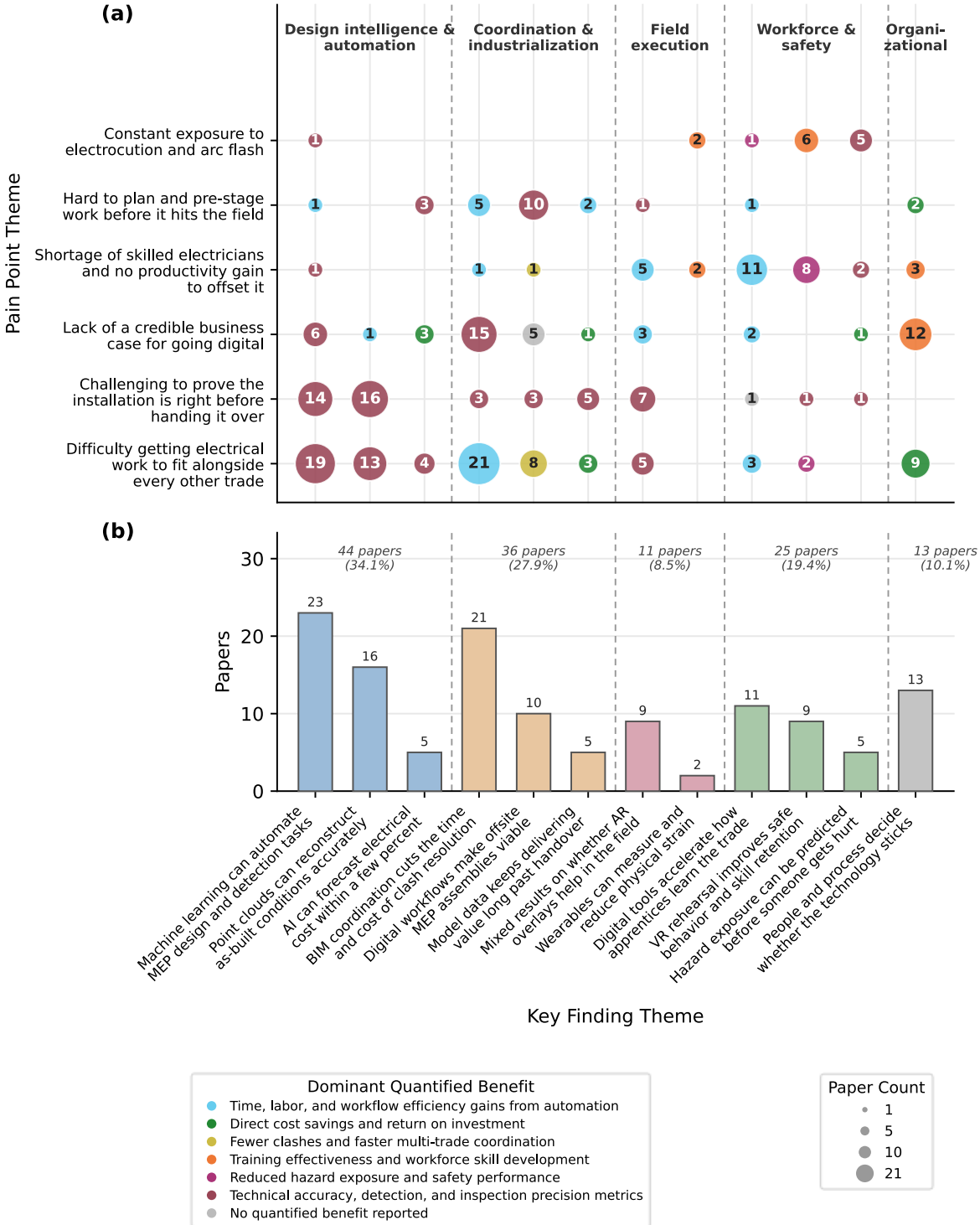


Figure 7. Alignment of pain points, key findings, and quantified benefits across the 129 reviewed studies.

The sparser cells are more revealing than the dense ones. Pain points tied to workforce, safety, and the business case for digitization map to thin columns: wearables measuring physical strain appear in two papers, hazard-exposure prediction in five, and cost forecasting in five. Pooladvand et al. (2024) and Aibinu et al. (2015) stand almost alone in their cells, which means the questions an electrical contractor is most likely to ask, whether a tool will reduce injuries or justify its cost, rest on isolated studies rather than a converging body of evidence. The one substantial signal outside the technical core is the finding that people and process decide whether the technology sticks (13 papers), anchored by Dossick and Neff (2010), a caution about adoption rather than a demonstrated solution.

Panel b shows what the literature actually measures. Efficiency and workflow gains from automation dominate at 44 papers (34.1%), followed by direct cost savings and return on investment at 36 papers (27.9%). Both are legitimate, but read against panel a they largely restate the coordination and automation cluster in benefit terms. The outcomes most consequential to a contractor's liability and reputation, safety performance and quality or inspection precision, occupy the smallest segments, and eleven papers (8.5%) quantify no benefit at all. Even the strongest field-tested cost claims come from single-site work, such as the $9M and six-month savings reported by Khanzode et al. (2008), while Jang and Lee (2018) expose the opposite edge of the same coin, cutting on-site MEP duration from 36 to 3 days yet raising total labor by 13.5%. Chalhoub et al. (2021) and Gorski et al. (2025) likewise report mixed or context-bound gains. Figure 7 thus shows a literature confident where the problem is computationally convenient and quiet where the contractor's stakes are highest.

From Synthesis to Interface: An Interactive Web Application

The review was additionally operationalized as a public web application (https://review.electriai.com/), whose construction pipeline is summarized in Figure 8. The application reuses the evidence base assembled through the PRISMA process, moving from 1,085 records to the 172 screened on title and abstract and finally to the 129 studies coded across forty-four dimensions. Two artifacts are prepared offline and shipped to the browser: the structured records, reshaped into the views presented above, and a vector index encoding each paper as a 768-dimensional embedding. A static visualization pipeline filters and aggregates the coded records to render the figures and to drill from any chart into the underlying papers, while a retrieval-augmented pipeline embeds a user query, retrieves the most similar papers, and returns a cited response. Three public surfaces, an interactive Findings view, a searchable Papers database, and a retrieval-grounded assistant, expose the same set of studies through visual, lookup, and conversational access.

The application is a dissemination and transparency instrument rather than an independent contribution, and its limitations follow from that design. Because both the charts and the assistant draw exclusively on the 129 coded studies, the tool inherits every skew documented above, and a fluent conversational interface risks lending unwarranted authority to an evidence base that is BIM-centric, phase-concentrated, and thin on field-tested outcomes. Retrieval by embedding similarity can surface papers that are semantically close but contextually mismatched, and the index is fixed at the review's cutoff, making the deployment a snapshot that will drift unless periodically re-embedded. Finally, and consistent with the pattern this review criticizes, the application's usefulness for practitioner decision-making is asserted rather than demonstrated, since no usability evaluation with electrical contractors has been conducted.

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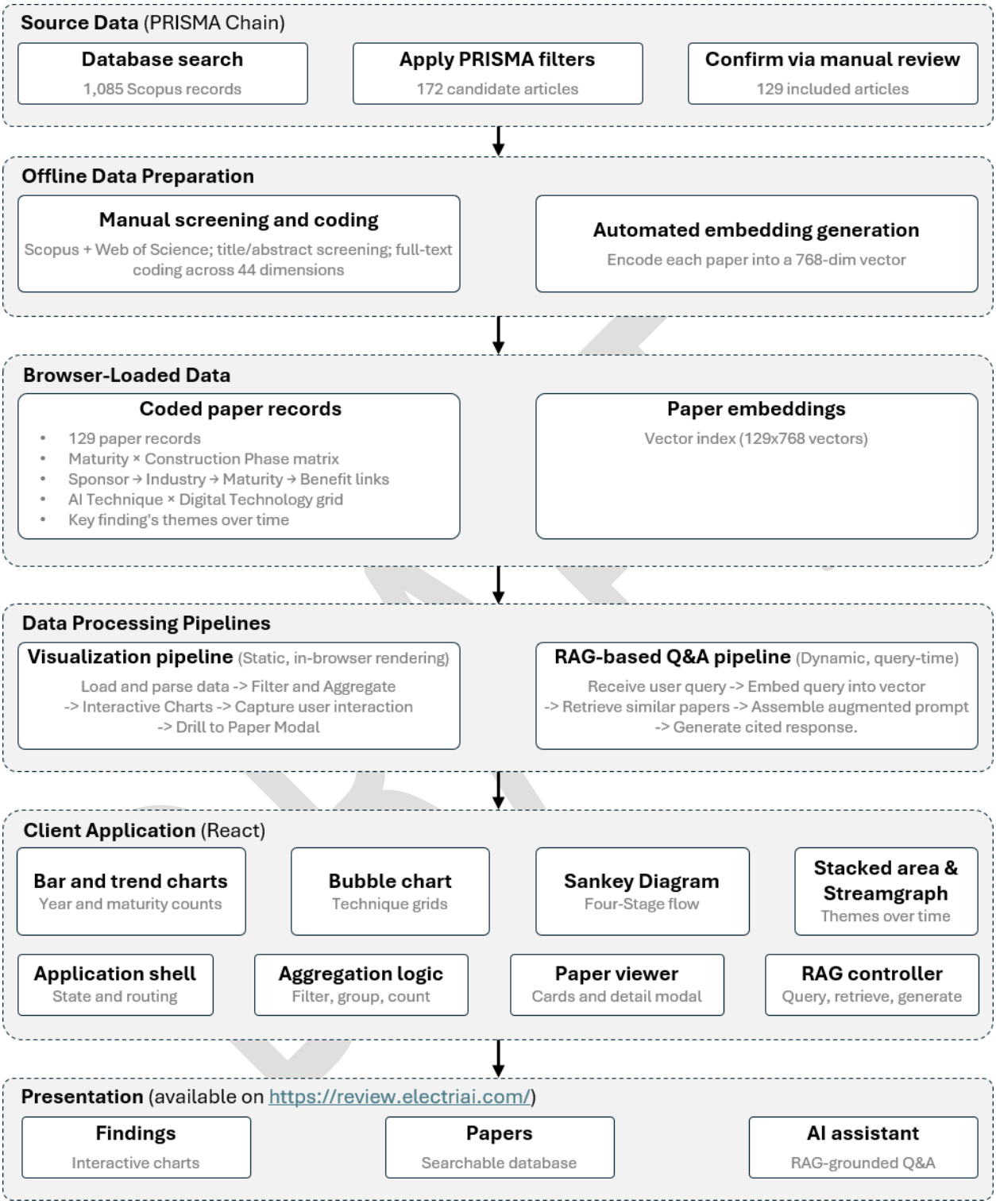


Figure 8. System architecture of the review web application, from the PRISMA-derived set of studies through offline embedding to browser-based visualization and retrieval-augmented querying (available at https://review.electriai.com/).

FUTURE RESEARCH AGENDA

The reviewed studies do not only report what has been done; most also state what should be done next. Read together, these self-declared agendas convert a scattered set of concluding paragraphs into a map of where the field believes its own frontier lies. Future-work statements were extracted verbatim from each paper and coded thematically at the paper level into the seven clusters summarized in Table 1. Because a single paper often proposes directions spanning several themes, papers were permitted to enter more than one cluster, and the cluster percentages therefore sum to well above 100%. All percentages are calculated against the full set of 129 papers and so express the share of the reviewed literature endorsing a given direction rather than a partition of it. Seven papers contribute to no cluster and remain in the denominator, making each reported percentage a conservative estimate of thematic prevalence.

What the Field Asks of Itself

The clusters are uneven in weight, and that unevenness is itself a finding. Technical Performance and Algorithmic Refinement (C3) is the largest at 43.4% (n = 56), followed by Cross-Technology and System Integration (C4) at 38.8% (n = 50) and Workforce Training, Usability, and Human-Centered Adoption (C6) at 31.8% (n = 41). The field's stated priorities thus skew toward improving and connecting the tools themselves. By contrast, Real-World and Real-Time Deployment Validation (C1, 23.3%) and Standardization, Cost Modeling, and Governance (C7, 20.9%), the two clusters most directly concerned with whether these tools work on an active jobsite and whether an electrical contractor can justify paying for them, are the two smallest. This ordering mirrors the maturity and industry-involvement patterns of Figures 3 and 6: a literature confident about algorithms and comparatively quiet about deployment continues to propose more algorithms. The agenda the field sets for itself would, if followed, reproduce the imbalance this review documents.

What the Field Does Not Ask For

Thematic coding of authors' own words privileges directions that are stated explicitly, so the silences must be inferred from the reviewed studies rather than read from Table 1, and four are conspicuous. First, no cluster treats the electrical trade as a distinct unit of analysis; future work is proposed for MEP or for construction generally, perpetuating the aggregation this review set out to disentangle. Second, the Execution phase, where most electrical rework originates (Figure 5), attracts almost no forward-looking attention despite being the phase in which errors remain preventable rather than merely recorded. Third, code compliance appears only as a downstream application of interpretability methods within C7, never as a research problem in its own right, even though it is among the most codified and therefore most automatable tasks an electrical contractor performs. Fourth, energy and carbon performance are absent from the stated agenda entirely, a striking omission for a trade whose installed systems govern operational building energy use.

Six Priority Directions for Electrical Construction Research

Reading the field's stated agenda against the gaps documented in the Results suggests six priorities that would move the evidence base toward the decisions electrical contractors actually face. First, LLM-assisted code compliance: automated parsing of the National Electrical Code and local amendments against model or as-built data, evaluated on documented compliance outcomes rather

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than benchmark accuracy. Second, AI-driven electrical cost estimation and business-case modeling, validated on completed projects rather than analytical estimates, which addresses the C7 gap directly. Third, execution-phase productivity and quality monitoring, including wearable and environmental sensing integrated with installation tracking, extending the isolated work of Pooladvand et al. (2024) into a coherent thread. Fourth, field-validation protocols for point cloud and computer vision methods that report performance degradation under occlusion, congestion, and variable lighting alongside deployment and integration cost. Fifth, organizational and contractual research on adoption, taking up the line Dossick and Neff (2010) opened and the reviewed literature has largely left unexamined. Sixth, open, trade-specific datasets and benchmarks for electrical components and installation states, without which the generalizability called for in C2 cannot be tested. Each priority is defined by an outcome the literature does not currently report, not by a technique it has yet to try. Across all six, the next logical stage is primary empirical research undertaken with electrical contractors on live projects. The most promising candidates, LLM-assisted code compliance, AI-supported cost estimation, execution-phase productivity monitoring, and automated inspection, should be tested through pilot deployments with baseline or comparison conditions and measured on contractor-relevant outcomes, including productivity, cost, labor hours, rework, safety, quality, decision-making, and return on investment. Such studies would move the research program from asking whether AI can perform a task to demonstrating whether it creates measurable value for electrical contractors and project performance.

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506 **Table 1. Synthesis of future research directions proposed across 129 reviewed AI-in-**
507 **electrical-construction studies, grouped into thematic clusters.**

Future Direction Cluster	Directions Proposed	Representative Studies
C1. Real-World & Real-Time Deployment Validation (n = 30, 23.3%). Field trials, live jobsite testing, empirical impact measurement	(1) Move from simulated or retrospective evaluation to live field trials on active jobsites. (2) Validate tools and frameworks across multiple case-study projects rather than a single site. (3) Measure impact empirically (time, cost, error rate) rather than through analytical or perception-based estimates. (4) Test AR/MR and sensing tools under live conditions with experienced practitioners.	Girgin et al. (2023); Alruwaythi & Goodrum (2019); Deng et al. (2025); Bashir et al. (2024); Chalhoub et al. (2021)
C2. Dataset Expansion, Diversity & Generalizability (n = 32, 24.8%). Larger and broader samples, synthetic augmentation, balanced datasets	(1) Expand datasets to cover more projects, building types, and regulatory or geographic contexts. (2) Collect larger, more balanced samples across practitioner experience levels and project outcomes. (3) Use synthetic or augmented data to fill gaps in underrepresented classes. (4) Broaden annotation coverage to component types omitted from current training data.	Yung et al. (2014); Michaud et al. (2019); Yue et al. (2025); Yin et al. (2025); Urbieta et al. (2026); Wang & Fang (2025); Lines and Smithwick (2019)
C3. Technical Performance & Algorithmic Refinement (n = 56, 43.4%). Robustness, accuracy, efficiency, advanced modeling (generative AI, LLMs, RL, graph learning)	(1) Improve model robustness to occlusion, lighting variation, and complex backgrounds. (2) Explore generative models, LLMs, reinforcement learning, and graph-based learning beyond current baselines. (3) Reduce computational cost through lightweight design, pruning, or distillation for site-deployable inference. (4) Reduce reliance on manual parameter tuning and prior domain knowledge.	Nazmul Huda et al. (2012); Yan et al. (2024); Deng et al. (2025); Sun et al. (2026); Jing et al. (2025); Yue et al. (2025)
C4. Cross-Technology & System Integration (n = 50, 38.8%). BIM, digital twin, and IoT coupling; unified pipelines; automated coordination workflows	(1) Couple BIM with digital twin, IoT, and real-time sensor streams for adaptive, closed-loop workflows. (2) Link siloed platforms and point solutions into a single interoperable pipeline. (3) Automate clash detection and resolution end-to-end rather than treating detection as an endpoint. (4) Extend rule-based routing engines further into design and coordination to reduce manual 2D steps.	Lu & Wong (2018); Yan et al. (2024); Deng et al. (2025); Alshboul & Shehadeh (2026)
C5. Scope & Functional Extension to New Contexts (n = 37, 28.7%). Extension to other trades, building types, components, or regulatory contexts	(1) Extend proposed methods beyond their original trade, building type, or component set. (2) Apply approaches validated in one region or code context to other jurisdictions and standards. (3) Broaden system functionality to less common or more complex conditions outside the initial scope.	Lee et al. (2014); Yin et al. (2025); Alshboul & Shehadeh (2026); Urbieta et al. (2026); Dallasega et al. (2023)
C6. Workforce Training, Usability & Human-Centered Adoption (n = 41, 31.8%). Usability testing, gamified and XR training, competency standards, trust and adoption	(1) Conduct usability studies with practitioners across experience levels, not only experts. (2) Investigate gamification and VR/AR training to strengthen engagement and skill transfer. (3) Study organizational and stakeholder adoption dynamics, including trust and cross-discipline collaboration. (4) Develop competency standards and certified training tied to new tools.	Tanaka et al. (2023); Alruwaythi & Goodrum (2019); Sun et al. (2026); Uhm et al. (2017); Dossick and Neff (2010)
C7. Standardization, Cost Modeling & Governance (n = 27, 20.9%). Formal standards, empirical cost validation, data governance, interpretability and compliance	(1) Develop formal standards for emerging practices such as prefabrication and BIM-based coordination. (2) Empirically validate cost and benefit models on real projects rather than theoretical estimates. (3) Investigate governance, data-sharing, and cloud collaboration frameworks across stakeholders. (4) Extend interpretability and rule-based reasoning to support compliance and auditability.	Yan et al. (2024); Deng et al. (2025); Xie et al. (2022); Daszczyński et al. (2022); Akponeware and Adamu (2017)

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CONCLUSION

This review synthesized 129 peer-reviewed studies to establish what is known, and what is only claimed, about AI and digital technologies in electrical construction. Three findings hold across the six analytical views presented. First, the evidence base is BIM-centric and phase-concentrated: BIM anchors the keyword network and appears in roughly half of the reviewed studies, while research clusters in preconstruction coordination and closeout documentation and thins to near-absence at execution, the phase in which most electrical rework originates. Second, the field is growing while its empirical grounding weakens; publication volume more than tripled after 2021 as the field-tested share fell from above 75% to between 29% and 45%, and quantified benefits are dominated by algorithm performance rather than by cost, safety, or productivity outcomes. Third, the electrical trade remains subordinated to the MEP aggregate, and industry engagement is shallow: a third of studies report no funding source, and only twelve acknowledge ELECTRI International or affiliated support.

The contribution of this review is not a claim that more research is needed but a specification of which research would change practice. The taxonomy organized by the electrical contractor value chain, the sponsorship and maturity profile traced in Figure 6, and the six priorities set out above together identify where the literature's confidence is warranted and where it is not. For electrical contractors, the practical implication is that published accuracy figures should be read as upper bounds established under favorable conditions until field-validation and deployment-cost evidence catches up. For researchers and funders, the implication is that the marginal value of another lab-validated model is now low relative to a single well-documented field deployment reporting cost, integration burden, and failure modes.

The review is bounded by its sources and its method. Coverage is limited to English-language, peer-reviewed journal articles and reviews indexed in Scopus and Web of Science, which excludes practitioner reports, standards documents, and vendor evidence that may carry deployment information the academic literature lacks. The search closed on 23 April 2026, so 2026 is a partial publication year and its counts are not comparable with those of complete years. Coding was performed against a single common frame, and both the thematic synthesis and the future-work clusters reflect interpretive judgment applied consistently rather than statistically verified inter-rater agreement. These constraints bound the precision of the percentages reported here; given the consistency of the pattern across independent analytical views, they do not bound the direction of the conclusion.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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APPENDIX A

Table A1. Search string applied to titles, abstracts, and keywords on 23 April 2026.

Block	Operator	Search terms
1. Electrical trade		"electrical construction" OR "electrical contractor*" OR "electrical installation*" OR "electrical trade" OR "electrical works" OR "electrical subcontractor*" OR "electrical fit-out" OR "electrical wiring" OR "electrician*" OR ("MEP" AND "electrical") OR ("building services" AND "electrical") OR "Division 26" OR "NECA" OR "ELECTRI International"
2. AI and digital technologies	AND	"artificial intelligence" OR "machine learning" OR "deep learning" OR "neural network*" OR "computer vision" OR "natural language processing" OR "reinforcement learning" OR "generative AI" OR "large language model*" OR "expert system*" OR "fuzzy logic" OR "genetic algorithm*" OR "building information model*" OR "BIM" OR "digital twin*" OR "Internet of Things" OR "IoT" OR "virtual reality" OR "augmented reality" OR "mixed reality" OR "extended reality" OR "wearable*" OR "exoskeleton*" OR "robot*" OR "automation" OR "drone*" OR "unmanned aerial vehicle*" OR "UAV" OR "laser scanning" OR "point cloud" OR "sensor*" OR "blockchain" OR "digitalization" OR "digitalisation" OR "Industry 4.0" OR "Construction 4.0" OR "digital transformation"
3. Non-construction electrical contexts	AND NOT	"smart grid*" OR "power grid" OR "microgrid*" OR "electric vehicle*" OR "EV charging" OR "photovoltaic*" OR "solar panel*" OR "wind turbine*" OR "smart meter*" OR "load forecasting" OR "harmonic distortion" OR "substation protection" OR "transmission line*" OR "appliance identification" OR "demand forecasting" OR "power electronics"
Limits		Publication years 2006 to the search date; English language; journal articles and reviews.

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